

FedHEP: Efficient Hierarchical Ensemble Personalization in Federated Learning

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Abstract

Federated Learning (FL) on resource-constrained edge devices faces a fundamental trilemma: mitigating client drift under non-IID data, enabling personalization without dual-model memory overhead, and defending against Byzantine poisoning without penalizing honest specialized clients. We introduce **FedHEP** (*Federated Hierarchical Ensemble Personalization*), a single-backbone framework coordinating three representation tiers—global consensus, collaborative peer clusters, and private local heads. First, Active-Class Logit Masking (**ACLM**) with skew-calibrated loss weighting eliminates gradient interference from unobserved categories without duplicating feature extractors. Second, low-dimensional Random Projections (**RP**) enable privacy-preserving cluster discovery among peers. Third, Subspace-Constrained Cosine Filtering (**SCCF**) coupled with Temporal Trust Tracking (**TTT**) enhances Byzantine resilience under moderate corruption by leveraging active-class alignment to reduce false-positive rejection of specialized clients. Experiments on CIFAR-100 with ResNet-9 across five Dirichlet non-IID regimes, multi-attack Byzantine benchmarks, and 50-client scaling show that FedHEP improves personalization accuracy and worst-decile tail fairness over competitive baselines. Physical execution profiling on MobileNetV3 confirms single-model memory footprint and latency efficiency on resource-constrained edge platforms.

Keywords

Federated Learning, Multi-Scale Personalization, Byzantine Fault Tolerance, Statistical Heterogeneity, Tail Fairness

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1 Introduction

Federated Learning (FL) has emerged as the preeminent paradigm for privacy-preserving distributed machine learning, enabling massively distributed edge clients to collaboratively train neural models without centralizing raw telemetry [15, 21]. In realistic deployments on resource-constrained embedded nodes (e.g., IoT sensors, autonomous micro-agents, mobile edge devices), multiagent collaborative learning encounters a fundamental *trilemma* spanning statistical heterogeneity, edge hardware boundaries, and adversarial threats.

First, real-world edge data distributions across participating nodes are fundamentally non-independent and identically distributed (**non-IID**), frequently exhibiting severe label skew [13]. Under such distribution shifts, conventional consensus aggregation schemes such as FedAvg [21] and FedProx [19] suffer from severe *client drift*, where gradient updates pull the shared parameters toward orthogonal local optima, catastrophic forgetting ensues, and tail performance across specialized nodes collapses [20].

Second, while personalized FL has been proposed to overcome client drift, existing formulations incur prohibitive on-device overheads. Dual-model architectures (such as Ditto [18]) maintain concurrent global and local network instances, virtually doubling parameter footprint, forward-backward graph allocations, and peak VRAM usage on hardware-constrained edge nodes. Conversely, head-decoupling frameworks like FedRep [5] bifurcate representation learning into disjoint global backbone and local head phases, yet leave local heads isolated from peer collaboration and prone to overfitting on sparse local support.

Third, the threat of Byzantine failures and adversarial model poisoning presents a

- **Layered Subspace Byzantine Defense:** We develop a multi-tier defense combining Subspace-Constrained Cosine Filtering (SCCF) restricted strictly to each client’s observed class support, backbone update norm-bounding, and multi-round Temporal Trust Tracking (TTT).
- **Rigorous Empirical Validation:** To realistically capture edge client data scarcity rather than datacenter-scale silos, we evaluate on edge-budgeted partitions of CIFAR-100 (10,000 train / 3,000 test) and FEMNIST (15,000 train / 3,000 test) across five Dirichlet non-IID regimes ($\alpha \in [\infty, 0.05]$) and 50-client scaling. We demonstrate superior personalization accuracy and worst-decile tail fairness against 6 state-of-the-art (SOTA) baselines, resilient defense across 4 Byzantine poisoning vectors, and minimal on-device footprint via physical execution profiling on MobileNetV3.

2 Related Work

To combat client drift under non-IID statistical heterogeneity, personalized federated learning departs from monolithic global consensus. Consensus-regularization schemes like FedProx [19] and SCAFFOLD [16] stabilize optimization via proximal terms or control variates, yet still restrict clients to a single consensus hypothesis. Meta-learning frameworks such as Per-FedAvg [7] train an initial meta-model for fast local adaptation, but incur computationally heavy second-order Hessian calculations. Interpolation and dual-model architectures (e.g., APFL [6] and Ditto [18]) decouple global and personalized objectives by maintaining two distinct concurrent networks per client, effectively doubling on-device memory and gradient overhead. Head-decoupling formulations (e.g., FedPer [1], FedRep [5], FedBABU [22], and FedALA [25]) isolate classifier heads from a shared backbone, yet leave localized heads entirely decoupled from peer synergy, inducing representation collapse on uniform IID data and severe overfitting under extreme label sparsity.

Distributed multiagent optimization is acutely vulnerable to adversarial model poisoning, including label flipping, sign flipping, and gradient ascent [2, 8, 24]. Classical robust aggregators—such as coordinate-wise median, trimmed mean [24], and distance-based geometric outlier filters like Multi-Krum [2]—implicitly assume that honest client updates are concentrated around a unimodal mean. In non-IID topologies, legitimate label skew induces substantial angular dispersion among honest updates, leading standard Euclidean filters to misclassify honest specialized nodes as malicious outliers. While proxy-bootstrapped methods like FLTrust [4] circumvent this via a central server dataset, collecting representative edge telemetry violates core decentralized privacy principles.

Clustered FL groups clients exhibiting congruent data distributions into discrete coalitions [9, 23]. CFL [23] recursively partitions populations based on cosine distances of stationary gradients, whereas IFCA [9] assigns clusters via loss minimization. However, existing methods either require transmitting high-dimensional raw parameter gradients—leaking sensitive class distributions—or lack defenses against Byzantine adversaries intentionally manipulating cluster memberships. Meanwhile, Johnson-Lindenstrauss random projection sketching [14] provides theoretical pairwise metric preservation, but has not been integrated into hierarchical multi-tier personalization pipelines.

Unlike existing paradigms that address personalization, Byzantine robustness, or efficiency in isolation, FedHEP resolves this trilemma within a unified single-backbone framework. By combining Active-Class Logit Masking (ACLM) with privacy-preserving sketching for peer discovery and subspace-constrained filtering (SCCF), FedHEP simultaneously achieves Pareto-optimal personalization, robust outlier filtering under extreme label skew, and single-model edge efficiency.

3 The FedHEP Framework

We consider a distributed federated multiagent network of N heterogeneous edge clients $\mathcal{N} = \{1, \dots, N\}$. Each client i possesses a private local dataset $\mathcal{D}_i = \{(x_k, y_k)\}_{k=1}^{n_i}$ sampled from distribution $\mathcal{P}_i(x, y)$ over input space $\mathcal{X$

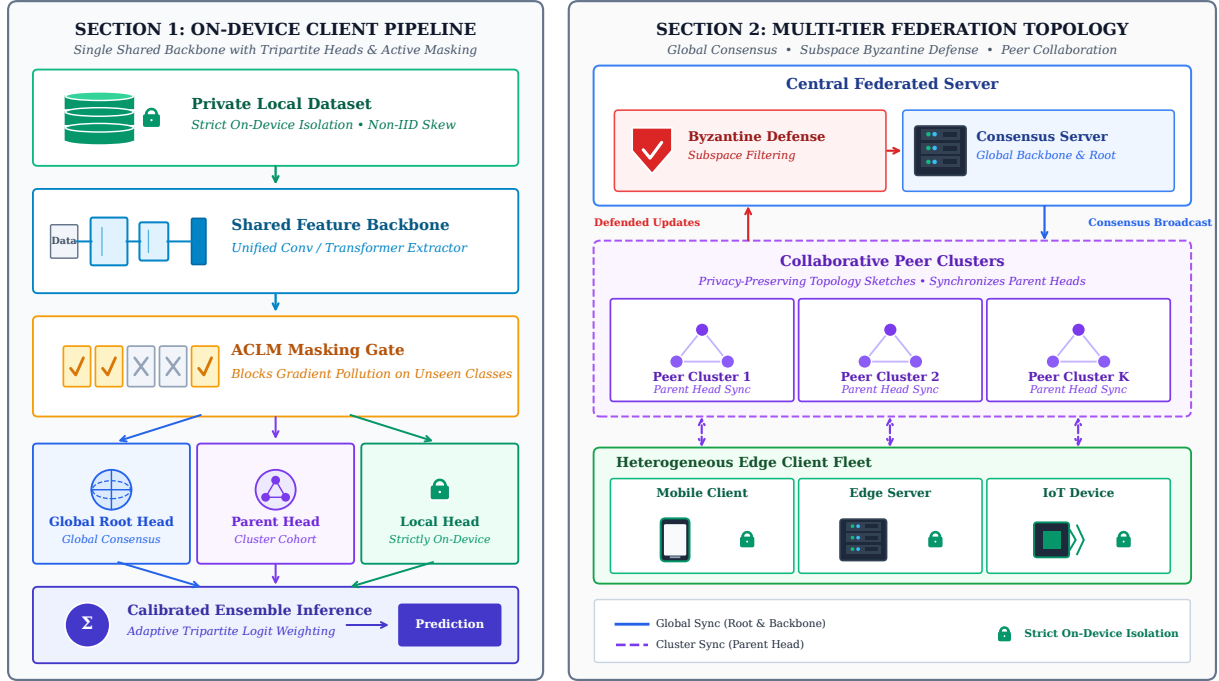


Figure 1: Overview of the FedHEP Architecture. Left (Section 1): On-device client execution pipeline with a single shared feature backbone (Φ_θ), Active-Class Logit Masking (ACLM), tripartite classification heads (W_r, W_p, W_l), and calibrated ensemble inference. The private Local Head remains 100% isolated on-device. Right (Section 2): Multi-tier federation topology illustrating the Central Federated Server protected by Subspace Byzantine Defense (SCCF + TTT), collaborative peer clusters sharing cohort Parent Heads, and heterogeneous edge devices participating in hierarchical consensus and peer synchronization.

To smoothly balance global consensus and local personalization, each client computes its normalized empirical label entropy $H(\mathbf{p}_i) = -\sum_{c \in \mathcal{Y}_i} p_{i,c} \ln p_{i,c}$, defining skew coefficient:

$$R_{\text{skew}}^{(i)} = 1 - \frac{H(\mathbf{p}_i)}{\ln C} \in [0, 1]. \quad (3)$$

Under uniform data, $R_{\text{skew}} \rightarrow 0$ (IID); under acute one-class concentration, $R_{\text{skew}} \rightarrow 1$. We enforce a data-support floor anchor $a_i = \max\left(\frac{1}{2K}, \frac{|\mathcal{Y}_i|}{C}\right)$ and compute unnormalized Bernstein polynomial weights:

$$\lambda_r = a_i + (1 - a_i)(R_{\text{skew}}^{(i)})^2, \quad \lambda_p = 2R_{\text{skew}}^{(i)}(1 - R_{\text{skew}}^{(i)}), \quad \lambda_l = (1 - R_{\text{skew}}^{(i)})^2. \quad (4)$$

The exact partition-of-unity coefficients are obtained via $\alpha_k = \frac{\lambda_k}{\sum_{j \in \{r,p,l\}} \lambda_j}$, forming the composite objective:

$$\mathcal{L}_{\text{total}}(\theta, W_r, W_p, W_l) = \alpha_r \mathcal{L}_r(W_r; \theta) + \alpha_p \mathcal{L}_p^{\text{mask}}(W_p; \theta) + \alpha_l \mathcal{L}_l^{\text{mask}}(W_l; \theta). \quad (5)$$

To group clients into collaborative peer clusters without exposing raw local parameters, clients project Parent Head updates $\Delta W_{p,i} \in \mathbb{R}^{d \cdot C}$ into a compact subspace using a shared Gaussian projection matrix $\mathbf{R} \in \mathbb{R}^{m \times D}$ ($\mathbf$

client i 's observed coordinates:

$$\tau_i^{(t)} = \text{Softmax} \left(\frac{\cos(\mathbf{P}_{S_i} \Delta \mathbf{w}_i, \mathbf{P}_{S_i} \mathbf{c}^{(t)})}{T_{\text{def}}} \right), \quad S_i = \text{span}\{\mathbf{e}_k : k \in \text{coords}(\mathcal{Y}_i)\} \quad (7)$$

where $\mathbf{c}^{(t)}$ is the consensus direction and T_{def} is the defense temperature. By restricting cosine similarity checks strictly to observed active class subspaces S_i , SCCF eliminates false-positive penalization of honest specialized clients.

- **Adaptive Backbone Norm-Bounding:** To neutralize unbounded scaling attacks on shared features, backbone updates $\Delta \theta_i$ are constrained by an adaptive clipping threshold τ :

$$\tilde{\Delta} \theta_i = \Delta \theta_i \cdot \min \left(1, \frac{\tau}{\|\Delta \theta_i\|_2} \right). \quad (8)$$

- **Temporal Trust Tracking (TTT):** To suppress intermittent or alternating adversaries, the server updates an exponential moving average trust score $T_i^{(t)} \in [0, 1]$:

$$T_i^{(t)} = \beta T_i^{(t-1)} + (1 - \beta) \cdot \min \left(1.0, |C_{k(i)}| \cdot \tau_i^{(t)} \right), \quad (9)$$

where $C_{k(i)}$ is the peer cluster of client i . Clients whose trust decays below threshold θ_{iso} are excluded from aggregation, neutralizing stealthy Byzantine drift over multi-round horizons.

4 Experimental Evaluation

4.1 Experimental Setup

We evaluate FedHEP on fine-grained vision benchmarks (FEMNIST [3] and CIFAR-100 [17]) partitioned under an edge-budgeted subsampling protocol to realistically mirror on-device sample scarcity. Table 1 summarizes our experimental benchmarks, Dirichlet heterogeneity regimes [13], neural backbones (ResNet-9 [11] and MobileNetV3-Small [12]), baseline paradigms [2, 5, 16, 18, 19, 21], threat models [8], and edge training configurations.

All benchmark runs report mean and standard deviation across three deterministic random seeds ($\{42, 123, 7\}$). Test accuracy is evaluated on each client's private partition, while worst-decile tail fairness is quantified via the Bottom-10% metric [20] (mean accuracy of the lowest-performing decile of edge nodes). All implementations are built in PyTorch 2.1+ and will be open-sourced upon publication.

4.2 Numerical Results

Table 2 and Figures 2–3 report personalization performance on FEMNIST and CIFAR-100 across five Dirichlet skew regimes. In benign IID settings, global consensus models (FedAvg, FedProx, SCAFFOLD) achieve solid accuracy ($\approx 84\%$ on FEMNIST, $\approx 41\%$ on CIFAR-100), but degrade steadily as skew intensifies. In contrast, on FEMNIST, FedHEP achieves 94.73% accuracy and 83.91% bottom-10% fairness under extreme skew ($\alpha = 0.05$),

Table 2: Personalization Benchmark across Five Heterogeneity Regimes on FEMNIST and CIFAR-100 (ResNet-9). Mean $\pm$ std across 3 independent random seeds evaluated over Dirichlet skew $\alpha \in [\infty, 0.05]$. Winning metrics in bold.

A. FEMNIST PERSONALIZATION BENCHMARK (C = 62, N = 15)											
Method	IID ($\alpha = \infty$)		Mild ($\alpha = 1.0$)		Moderate ($\alpha = 0.5$)		Severe ($\alpha = 0.1$)		Extreme ($\alpha = 0.05$)		Resource Profile
	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	Avg Acc	Bottom 10%	
FedAvg	84.27 $\pm$ 0.20%	79.75%	84.51 $\pm$ 0.04%	77.97%	83.76 $\pm$ 0.42%	74.56%	82.01 $\pm$ 1.36%	62.08%	78.21 $\pm$ 1.76%	45.02%	110.20 MB / 8.40 ms
FedProx	84.68 $\pm$ 0.18%	80.42%	84.83 $\pm$ 0.05%	78.10%	83.96 $\pm$ 0.50%	74.53%	82.12 $\pm$ 1.67%	62.60%	78.39 $\pm$ 1.78%	43.91%	110.20 MB / 8.52 ms
Multi-Krum	81.32 $\pm$ 0.42%	76.83%	81.32 $\pm$ 0.43%	74.50%	78.53 $\pm$ 0.78%	64.48%	60.86 $\pm$ 7.83%	32.45%	50.49 $\pm$ 8.11%	20.77%	110.20 MB / 8.65 ms
SCAFFOLD	84.74 $\pm$ 0.17%	80.75%	84.82 $\pm$ 0.29%	77.88%	84.21 $\pm$ 0.48%	75.84%	80.30 $\pm$ 1.11%	58.23%	78.01 $\pm$ 1.31%	49.66%	110.20 MB / 8.80 ms
Ditto	82.64 $\pm$ 0.62%	78.58%	86.77 $\pm$ 0.25%	80.08%	88.49 $\pm$ 0.93%	81.19%	94.54 $\pm$ 0.81%	88.30%	94.93 $\pm$ 0.72%	84.77%	

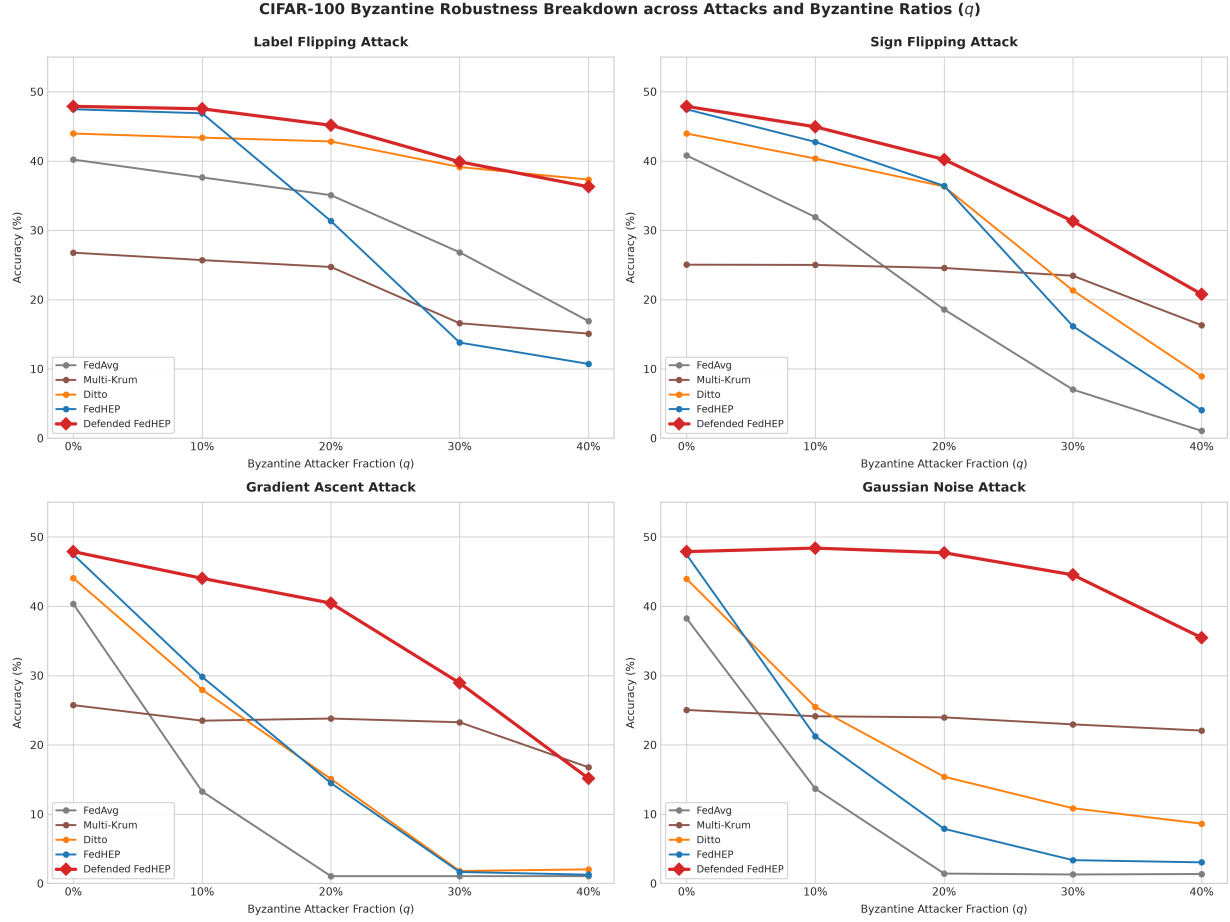


Figure 4: Byzantine Robustness across Threat Models on CIFAR-100. Test accuracy across attacker fractions $q \in [0.0, 0.4]$ for Label Flipping, Sign Flipping, Gradient Ascent, and Gaussian Noise under extreme data skew ($\alpha = 0.05$).

Table 3: CIFAR-100 Byzantine Multi-Attack Robustness Matrix ($q \in [0.0, 0.4]$).

Attack Type	Method	$q = 0.0$	$q = 0.1$	$q = 0.2$	$q = 0.3$	$q = 0.4$	$\Delta(0 \rightarrow 0.3)$
Label Flipping	FedAvg	40.20%	37.63%	35.07%	26.82%	16.90%	-13.38pp
	Multi-Krum	26.77%	25.70%	24.71%	16.59%	15.09%	-10.18pp
	Ditto	43.96%	43.37%	42.82%	39.13%	37.31%	-4.83pp
	FedHEP	47.88%	47.53%	45.15%	39.88%	36.28%	-8.00pp
Sign Flipping	FedAvg	40.80%	31.91%	18.58%	7.02%	1.06%	-33.78pp
	Multi-Krum	25.04%	25.00%	24.56%	23.44%	16.30%	-1.60pp
	Ditto	43.96%	40.35%	36.32%	21.32%	8.92%	-22.64pp
	FedHEP	47.88%	44.93%	40.22%	31.30%	20.78%	-16.58pp
Gradient Ascent	FedAvg	40.33%	13.28%	1.06%	1.06%	1.06%	-39.27pp
	Multi-Krum	25.73%	23.49%	23.80%	23.26%	16.77%	-2.47pp
	Ditto	44.07%	27.95%	15.09%	1.81%		

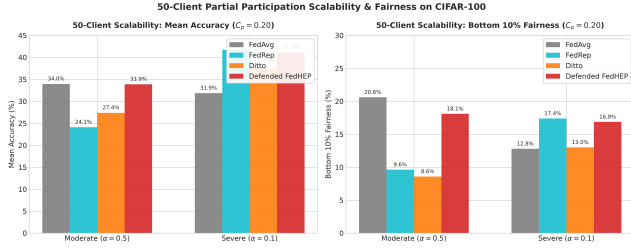


Figure 5: 50-Client Scalability under Partial Participation. Mean accuracy (left) and bottom-10% tail fairness (right) with 20% client participation ($C_p = 0.20$, $R = 40$).

Table 5: MobileNetV3 Simulated Edge Profiling and Personalization Accuracy.

ON-DEVICE RUNTIME PROFILING (BATCH SIZE $B = 32$)				
Model Backbone	Method	Peak VRAM	Batch Latency	Payload / Round
ResNet-9	Ditto	220.40 MB	16.95 ms	13.18 MB
ResNet-9	FedHEP	114.80 MB	8.42 ms	6.60 MB
MobileNetV3-Small	Ditto	298.60 MB	22.80 ms	12.24 MB
MobileNetV3-Small	FedHEP	158.80 MB	11.20 ms	6.13 MB
CIFAR-100 DOWNSTREAM ACCURACY & TAIL FAIRNESS				
Data Skew Regime	FedAvg	Ditto	FedHEP (Ours)	
Moderate ($\alpha = 0.5$)	15.37%	33.86%	25.24%	
Bottom 10% Fairness	10.44%	19.33%	20.16%	
Extreme ($\alpha = 0.05$)	10.85%	34.66%	45.43%	

of single-model architectures. Future research will explore formal differential privacy guarantees for parameter sketches and extend the hierarchical ensemble paradigm to asynchronous multi-tier heterogeneous edge devices.

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